

Human review packet: LBG24SpatialUnderreporting

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: LBG24SpatialUnderreporting

Paper id: LBG24SpatialUnderreporting

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Source URL: [official source](#)

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) AppliedModelingLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-repository-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper LBG24SpatialUnderreporting --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local reviewer-owned review record.

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Governing Lean declarations

These exact graph-bound declaration bodies are retained by the displayed specifications or reviewed prerequisites. They are supporting context and do not add source-result rows or semantic judgments.

AppliedModelingLib.Probability.Exponential.Model

Declaration location: reusable library

Lean declaration body

```
field rate:
AppliedModelingLib.Probability.Exponential.Model → ℝ

field rate_pos:
∀ (self : AppliedModelingLib.Probability.Exponential.Model), 0 < self.rate
```

AppliedModelingLib.Probability.Exponential.Model.measure

Declaration location: reusable library

Lean declaration body

```
type:
AppliedModelingLib.Probability.Exponential.Model → MeasureTheory.Measure ℝ

value:
fun (M : AppliedModelingLib.Probability.Exponential.Model) => ProbabilityTheory.expMeasure M.rate
```

Direct retained dependencies

- [AppliedModelingLib.Probability.Exponential.Model](#)

AppliedModelingLib.Probability.PoissonCountingMeasureByLaw

Declaration location: reusable library

Lean declaration body

```
field isProbability:
∀ {Ω : Type u_1} {X : Type u_2} [inst : MeasurableSpace Ω] [inst_1 : MeasurableSpace X] {P : MeasureTheory.Measure Ω}
{intensity : MeasureTheory.Measure X}
(self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity),
MeasureTheory.IsProbabilityMeasure P

field finiteCount:
{Ω : Type u_1} →
{X : Type u_2} →
[inst : MeasurableSpace Ω] →
[inst_1 : MeasurableSpace X] →
{P : MeasureTheory.Measure Ω} →
{intensity : MeasureTheory.Measure X} →
AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity → Set X → Ω → ℕ

field finiteCount_measurable:
∀ {Ω : Type u_1} {X : Type u_2} [inst : MeasurableSpace Ω] [inst_1 : MeasurableSpace X] {P : MeasureTheory.Measure Ω}
{intensity : MeasureTheory.Measure X}
(self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity) (s : Set X),
Measurable (self.finiteCount s)

field finiteCount_empty_ae:
∀ {Ω : Type u_1} {X : Type u_2} [inst : MeasurableSpace Ω] [inst_1 : MeasurableSpace X] {P : MeasureTheory.Measure Ω}
{intensity : MeasureTheory.Measure X}
(self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity),
∀m (ω : Ω) ∂P, self.finiteCount ∅ ω = 0

field finiteCount_mono:
∀ {Ω : Type u_1} {X : Type u_2} [inst : MeasurableSpace Ω] [inst_1 : MeasurableSpace X] {P : MeasureTheory.Measure Ω}
{intensity : MeasureTheory.Measure X}
(self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity) (s t : Set X),
MeasurableSet s →
MeasurableSet t → intensity s ≠ T → intensity t ≠ T → s ⊆ t → ∀ (ω : Ω), self.finiteCount s ω ≤ self.finiteCount t ω

field finiteCount_union_ae:
∀ {Ω : Type u_1} {X : Type u_2} [inst : MeasurableSpace Ω] [inst_1 : MeasurableSpace X] {P : MeasureTheory.Measure Ω}
{intensity : MeasureTheory.Measure X}
(self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity) (s t : Set X),
MeasurableSet s →
MeasurableSet t →
intensity s ≠ T →
intensity t ≠ T →
Disjoint s t → ∀m (ω : Ω) ∂P, self.finiteCount (s ∪ t) ω = self.finiteCount s ω + self.finiteCount t ω
```

```

field finiteCount_independent:
  ∀ {Ω : Type u_1} {X : Type u_2} [inst : MeasurableSpace Ω] [inst_1 : MeasurableSpace X] {P : MeasureTheory.Measure Ω}
    {intensity : MeasureTheory.Measure X}
    (self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity) (n : ℕ) (s : Fin n → Set X),
    (∀ (i : Fin n), MeasurableSet (s i)) →
    (∀ (i : Fin n), intensity (s i) ≠ ⊤) →
    (Pairwise fun (i j : Fin n) => Disjoint (s i) (s j)) →
    ProbabilityTheory.IndepFun (fun (i : Fin n) => self.finiteCount (s i)) P

```

```

field finiteCount_hasLaw:
  ∀ {Ω : Type u_1} {X : Type u_2} [inst : MeasurableSpace Ω] [inst_1 : MeasurableSpace X] {P : MeasureTheory.Measure Ω}
    {intensity : MeasureTheory.Measure X}
    (self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity) (s : Set X),
    MeasurableSet s →
    intensity s ≠ ⊤ →
    ProbabilityTheory.HasLaw (self.finiteCount s) (ProbabilityTheory.poissonMeasure (intensity s).toNNReal) P

```

AppliedModelingLib.Probability.PoissonProcess.ObservationWindow

Declaration location: reusable library

Lean declaration body

```

field startTime:
  AppliedModelingLib.Probability.PoissonProcess.ObservationWindow → ℝ

field endTime:
  AppliedModelingLib.Probability.PoissonProcess.ObservationWindow → ℝ

field start_le_end:
  ∀ (self : AppliedModelingLib.Probability.PoissonProcess.ObservationWindow), self.startTime ≤ self.endTime

```

AppliedModelingLib.Probability.PoissonProcess.ObservationWindow.exposure

Declaration location: reusable library

Lean declaration body

```

type:
  AppliedModelingLib.Probability.PoissonProcess.ObservationWindow → ℝ

value:
  fun (W : AppliedModelingLib.Probability.PoissonProcess.ObservationWindow) => W.endTime - W.startTime

```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow](#)

AppliedModelingLib.Probability.PoissonProcess.RateIndependent

Declaration location: reusable library

Lean declaration body

```

type:
  (ℝ → ℝ) → Prop

value:
  fun (f : ℝ → ℝ) => ∃ (c : ℝ), ∀ (rate : ℝ), f rate = c

```

AppliedModelingLib.Probability.PoissonProcess.countLikelihood

Declaration location: reusable library

Lean declaration body

```

type:
  ℝ → ℝ → ℕ → ℝ

value:
  fun (rate exposure : ℝ) (n : ℕ) => Real.exp (-(rate * exposure)) * (rate * exposure) ^ n / ↑n.factorial

```

AppliedModelingLib.Probability.PoissonProcess.cumulativeArrivalVector

Declaration location: reusable library

Lean declaration body

```
type:
(q : ℕ) → (Fin q → ℝ) → Fin q → ℝ

value:
fun (q : ℕ) (a : Fin q → ℝ) (a_1 : Fin q) => ∑ j : { j : Fin q // j ≤ a_1 }, a ↑j
```

AppliedModelingLib.Probability.PoissonProcess.exponentialBlockDensity

Declaration location: reusable library

Lean declaration body

```
type:
ℝ → (q : ℕ) → (Fin q → ℝ) → ENNReal

value:
fun (rate : ℝ) (q : ℕ) (a : Fin q → ℝ) => ∏ i : Fin q, ProbabilityTheory.exponentialPDF rate (a i)
```

AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint

Declaration location: reusable library

Lean declaration body

```
type:
ℝ → (ℕ → ℝ) → ℕ → ℝ

value:
fun (start : ℝ) (jumpTime : ℕ → ℝ) (a : ℕ) =>
  match a with
  | 0 => start
  | i.succ => jumpTime i
```

AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline

Declaration location: reusable library

Lean declaration body

```
field window:
AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline →
AppliedModelingLib.Probability.PoissonProcess.ObservationWindow

field count:
AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline → ℕ

field jumpTime:
AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline → ℕ → ℝ

field endpoint_mono:
∀ (self : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline),
  Monotone (AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint self.window.startTime self.jumpTime)

field last_le_end:
∀ (self : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline),
  AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint self.window.startTime self.jumpTime self.count ≤
  self.window.endTime
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow](#)
- [AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint](#)

AppliedModelingLib.Probability.PoissonProcess.interarrivalGapFromJumpTimes

Declaration location: reusable library

Lean declaration body

```
type:
ℝ → (ℕ → ℝ) → ℕ → ℝ

value:
fun (start : ℝ) (jumpTime : ℕ → ℝ) (i : ℕ) =>
  AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint start jumpTime (i + 1) -
  AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint start jumpTime i
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint](#)

AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.gap

Declaration location: reusable library

Lean declaration body

```
type:
(T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) → Fin T.count → ℝ

value:
fun (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) (j : Fin T.count) =>
  AppliedModelingLib.Probability.PoissonProcess.interarrivalGapFromJumpTimes T.window.startTime T.jumpTime ↑j
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.interarrivalGapFromJumpTimes](#)

AppliedModelingLib.Probability.PoissonProcess.noArrivalProb

Declaration location: reusable library

Lean declaration body

```
type:
ℝ → ℝ → ℝ

value:
fun (rate exposure : ℝ) => Real.exp (-(rate * exposure))
```

AppliedModelingLib.Probability.PoissonProcess.poissonRateLogLikelihoodKernel

Declaration location: reusable library

Lean declaration body

```
type:
ℕ → ℝ → ℝ → ℝ

value:
fun (count : ℕ) (exposure rate : ℝ) => ↑count * Real.log rate - rate * exposure
```

AppliedModelingLib.Probability.PoissonProcess.terminalTailFromJumpTimes

Declaration location: reusable library

Lean declaration body

```
type:
ℝ → ℝ → (ℕ → ℝ) → ℕ → ℝ

value:
fun (start endTime : ℝ) (jumpTime : ℕ → ℝ) (count : ℕ) =>
  endTime - AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint start jumpTime count
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint](#)

AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.tail

Declaration location: reusable library

Lean declaration body

```
type:
AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline → ℝ

value:
fun (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) =>
  AppliedModelingLib.Probability.PoissonProcess.terminalTailFromJumpTimes T.window.startTime T.window.endTime T.jumpTime
  T.count
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.terminalTailFromJumpTimes](#)

AppliedModelingLib.Probability.PoissonProcess.zeroInflatedCountLikelihood

Declaration location: reusable library

Lean declaration body

```
type:
ℝ → ℝ → ℝ → ℕ → ℝ

value:
fun (γ rate exposure : ℝ) (count : ℕ) =>
  if count = 0 then γ + (1 - γ) * AppliedModelingLib.Probability.PoissonProcess.countLikelihood rate exposure 0
  else (1 - γ) * AppliedModelingLib.Probability.PoissonProcess.countLikelihood rate exposure count
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.countLikelihood](#)

LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.firstReportSet

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ × ℝ → Prop

value:
fun (a b : ℝ) (a_1 : ℝ × ℝ) => ((fun (z : ℝ × ℝ) => z.1 + z.2)⁻¹) Set.loc a b) a_1
```

LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.calendarCount

Declaration location: paper-local

Lean declaration body

```
type:
{Ω : Type u_1} →
[inst : MeasurableSpace Ω] →
{P : MeasureTheory.Measure Ω} → LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P → NNReal → Ω → ℕ

value:
fun {Ω : Type u_1} [MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (M : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P) (t : NNReal) (a : Ω) =>
  M.markedBirths.finiteCount (LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.firstReportSet 0 ↑t) a
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonCountingMeasureByLaw](#)
- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel](#)
- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.firstReportSet](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalNoArrivalTail

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \text{ENNReal}$ 

value:
fun (rate a :  $\mathbb{R}$ ) => (Set.Ici 0).indicator (fun (tail :  $\mathbb{R}$ ) => ENNReal.ofReal (Real.exp (-(rate * tail)))) a
```

LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile

Declaration location: paper-local

Lean declaration body

```
field startDensity:
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T →  $\mathbb{R}$ 

field startDensity_nonnegative:
 $\forall \{T : \text{AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}\}$ 
  (self : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T),  $0 \leq \text{self.startDensity}$ 

field endKernel:
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T →
    (j : Fin (T.count + 1)) → ProbabilityTheory.Kernel (Fin  $\uparrow$  j →  $\mathbb{R}$ )  $\mathbb{R}$ 

field endKernel_isMarkov:
 $\forall \{T : \text{AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}\}$ 
  (self : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) (j : Fin (T.count + 1)),
    ProbabilityTheory.IsMarkovKernel (self.endKernel j)

field endDensity:
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T → (j : Fin (T.count + 1)) → (Fin  $\uparrow$  j →  $\mathbb{R}$ ) →  $\mathbb{R} \rightarrow \text{ENNReal}$ 

field endDensity_measurable:
 $\forall \{T : \text{AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}\}$ 
  (self : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) (j : Fin (T.count + 1)),
    Measurable (Function.uncurry (self.endDensity j))

field endKernel_eq_withDensity:
 $\forall \{T : \text{AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}\}$ 
  (self : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) (j : Fin (T.count + 1)),
    self.endKernel j = (ProbabilityTheory.Kernel.const (Fin  $\uparrow$  j →  $\mathbb{R}$ ) MeasureTheory.volume).withDensity (self.endDensity j)
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)

LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel

Declaration location: paper-local

Lean declaration body

```
field gamma:
  LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel →  $\mathbb{R}$ 

field gamma_nonnegative:
 $\forall$  (self : LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel),  $0 \leq \text{self.gamma}$ 

field gamma_le_one:
 $\forall$  (self : LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel),  $\text{self.gamma} \leq 1$ 

field rate:
  LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel →  $\mathbb{R}$ 

field rate_pos:
 $\forall$  (self : LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel),  $0 < \text{self.rate}$ 

field exposure:
  LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel →  $\mathbb{R}$ 

field exposure_pos:
 $\forall$  (self : LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel),  $0 < \text{self.exposure}$ 
```

LBG24SpatialUnderreporting.continuousDurationNoReportProbability

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow (\mathbb{R} \rightarrow \mathbb{R}) \rightarrow \mathbb{R}$ 

value:
fun (reportingRate :  $\mathbb{R}$ ) (durationDensity :  $\mathbb{R} \rightarrow \mathbb{R}$ ) =>
   $\int (t : \mathbb{R}) \text{ in } \text{Set.Ici } 0, \text{durationDensity } t * \text{AppliedModelingLib.Probability.PoissonProcess.noArrivalProb reportingRate } t$ 
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.noArrivalProb](#)

LBG24SpatialUnderreporting.continuousDurationFirstReportProbability

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow (\mathbb{R} \rightarrow \mathbb{R}) \rightarrow \mathbb{R}$ 

value:
fun (reportingRate :  $\mathbb{R}$ ) (durationDensity :  $\mathbb{R} \rightarrow \mathbb{R}$ ) =>
  1 - LBG24SpatialUnderreporting.continuousDurationNoReportProbability reportingRate durationDensity
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.continuousDurationNoReportProbability](#)

LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel.observedUniqueIncidentCount

Declaration location: paper-local

Lean declaration body

```
type:
{ $\Omega$  : Type u_1} →
[inst : MeasurableSpace  $\Omega$ ] →
{P : MeasureTheory.Measure  $\Omega$ } →
  LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel  $\Omega$  P →  $\mathbb{N} \rightarrow \Omega \rightarrow \mathbb{N}$ 

value:
fun { $\Omega$  : Type u_1} [MeasurableSpace  $\Omega$ ] {P : MeasureTheory.Measure  $\Omega$ }
(M : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel  $\Omega$  P) (n :  $\mathbb{N}$ ) ( $\omega$  :  $\Omega$ ) =>
 $\sum i \in \text{Finset.range } n, (M.\text{calendarFirstReports}.\text{calendarCount } (\uparrow i + 1) \omega - M.\text{calendarFirstReports}.\text{calendarCount } (\uparrow i) \omega)$ 
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.calendarCount](#)
- [LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel](#)

LBG24SpatialUnderreporting.continuousDurationNoReportProbabilityOfCumulativeIntensity

Declaration location: paper-local

Lean declaration body

```
type:
( $\mathbb{R} \rightarrow \mathbb{R}$ ) → ( $\mathbb{R} \rightarrow \mathbb{R}$ ) →  $\mathbb{R}$ 

value:
fun (cumulativeIntensity durationDensity :  $\mathbb{R} \rightarrow \mathbb{R}$ ) =>
   $\int (t : \mathbb{R}) \text{ in } \text{Set.Ici } 0, \text{durationDensity } t * \text{Real.exp } (-\text{cumulativeIntensity } t)$ 
```

LBG24SpatialUnderreporting.continuousDurationFirstReportProbabilityOfCumulativeIntensity

Declaration location: paper-local

Lean declaration body

```
type:
( $\mathbb{R} \rightarrow \mathbb{R}$ )  $\rightarrow$  ( $\mathbb{R} \rightarrow \mathbb{R}$ )  $\rightarrow$   $\mathbb{R}$ 

value:
fun (cumulativeIntensity durationDensity :  $\mathbb{R} \rightarrow \mathbb{R}$ ) =>
  1 -
    LBG24SpatialUnderreporting.continuousDurationNoReportProbabilityOfCumulativeIntensity cumulativeIntensity
    durationDensity
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.continuousDurationNoReportProbabilityOfCumulativeIntensity](#)

LBG24SpatialUnderreporting.continuousDurationObservedIncidentRateOfCumulativeIntensity

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow$  ( $\mathbb{R} \rightarrow \mathbb{R}$ )  $\rightarrow$  ( $\mathbb{R} \rightarrow \mathbb{R}$ )  $\rightarrow$   $\mathbb{R}$ 

value:
fun (incidentRate :  $\mathbb{R}$ ) (cumulativeIntensity durationDensity :  $\mathbb{R} \rightarrow \mathbb{R}$ ) =>
  incidentRate *
    LBG24SpatialUnderreporting.continuousDurationFirstReportProbabilityOfCumulativeIntensity cumulativeIntensity
    durationDensity
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.continuousDurationFirstReportProbabilityOfCumulativeIntensity](#)

LBG24SpatialUnderreporting.finiteGapPrefix

Declaration location: paper-local

Lean declaration body

```
type:
{count :  $\mathbb{N}$ }  $\rightarrow$  (Fin count  $\rightarrow$   $\mathbb{R}$ )  $\rightarrow$  (j : Fin (count + 1))  $\rightarrow$  Fin  $\uparrow$  j  $\rightarrow$   $\mathbb{R}$ 

value:
fun {count :  $\mathbb{N}$ } (gaps : Fin count  $\rightarrow$   $\mathbb{R}$ ) (j : Fin (count + 1)) (a : Fin  $\uparrow$  j) =>
  gaps ( $\uparrow$  a, LBG24SpatialUnderreporting.finiteGapPrefix._proof_1 j a)
```

LBG24SpatialUnderreporting.finiteArrivalPrefix

Declaration location: paper-local

Lean declaration body

```
type:
{count :  $\mathbb{N}$ }  $\rightarrow$  (Fin count  $\rightarrow$   $\mathbb{R}$ )  $\rightarrow$  (j : Fin (count + 1))  $\rightarrow$  Fin  $\uparrow$  j  $\rightarrow$   $\mathbb{R}$ 

value:
fun {count :  $\mathbb{N}$ } (gaps : Fin count  $\rightarrow$   $\mathbb{R}$ ) (j : Fin (count + 1)) (a : Fin  $\uparrow$  j) =>
  AppliedModelingLib.Probability.PoissonProcess.cumulativeArrivalVector ( $\uparrow$  j)
  (LBG24SpatialUnderreporting.finiteGapPrefix gaps j) a
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.cumulativeArrivalVector](#)
- [LBG24SpatialUnderreporting.finiteGapPrefix](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel

Declaration location: paper-local

Lean declaration body

```
field startWeight:
{count : ℕ} → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count → ENNReal

field endKernel:
{count : ℕ} →
  LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count →
  (j : Fin (count + 1)) → ProbabilityTheory.Kernel (Fin ↑ j → ℝ) ℝ

field endKernel_isMarkov:
∀ {count : ℕ} (self : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (j : Fin (count + 1)),
  ProbabilityTheory.IsMarkovKernel (self.endKernel j)

field endDensity:
{count : ℕ} →
  LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count →
  (j : Fin (count + 1)) → (Fin ↑ j → ℝ) → ℝ → ENNReal

field endDensity_measurable:
∀ {count : ℕ} (self : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (j : Fin (count + 1)),
  Measurable (Function.uncurry (self.endDensity j))

field endKernel_eq_withDensity:
∀ {count : ℕ} (self : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (j : Fin (count + 1)),
  self.endKernel j = (ProbabilityTheory.Kernel.const (Fin ↑ j → ℝ) MeasureTheory.volume).withDensity (self.endDensity j)

field stageSurvival_measurable:
∀ {count : ℕ} (self : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (i : Fin count),
  Measurable fun (gaps : Fin count → ℝ) =>
    (self.endKernel i.castSucc) (LBG24SpatialUnderreporting.finiteArrivalPrefix gaps i.castSucc) (Set.loi (gaps i))

field terminalDensity_measurable:
∀ {count : ℕ} (self : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count),
  Measurable fun (p : (Fin count → ℝ) × ℝ) =>
    self.endDensity (Fin.last count) (LBG24SpatialUnderreporting.finiteArrivalPrefix p.1 (Fin.last count)) p.2
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.finiteArrivalPrefix](#)

LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation.toDensityModel

Declaration location: paper-local

Lean declaration body

```
type:
{count : ℕ} →
  (M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count) →
  M.EndpointDensityPresentation → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count

value:
fun {count : ℕ} (M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count)
  (D : M.EndpointDensityPresentation) =>
  { startWeight := M.selectedStartLikelihood, endKernel := M.endpointResponse,
    endKernel_isMarkov := M.endpointResponse_isMarkov, endDensity := D.endpointDensity,
    endDensity_measurable := D.endpointDensity_measurable,
    endKernel_eq_withDensity := D.endpointResponse_eq_withDensity,
    stageSurvival_measurable := M.responseSurvival_measurable,
    terminalDensity_measurable := D.terminalDensity_measurable }
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.stageSurvival

Declaration location: paper-local

Lean declaration body

```
type:
{count : ℕ} → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count → (Fin count → ℝ) → Fin count → ENNReal

value:
fun {count : ℕ} (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (gaps : Fin count → ℝ)
  (i : Fin count) =>
  ((M.endKernel i.castSucc) (LBG24SpatialUnderreporting.finiteArrivalPrefix gaps i.castSucc)) (Set.loi (gaps i))
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.finiteArrivalPrefix](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalDensity

Declaration location: paper-local

Lean declaration body

```
type:
{count : ℕ} → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count → (Fin count → ℝ) × ℝ → ENNReal

value:
fun {count : ℕ} (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (p : (Fin count → ℝ) × ℝ) =>
  M.endDensity (Fin.last count) (LBG24SpatialUnderreporting.finiteArrivalPrefix p.1 (Fin.last count)) p.2
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.finiteArrivalPrefix](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.endpointWeight

Declaration location: paper-local

Lean declaration body

```
type:
{count : ℕ} → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count → (Fin count → ℝ) × ℝ → ENNReal

value:
fun {count : ℕ} (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (p : (Fin count → ℝ) × ℝ) =>
  (M.startWeight * ∏ i : Fin count, M.stageSurvival p.1 i) * M.terminalDensity p
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.stageSurvival](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalDensity](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalWeight

Declaration location: paper-local

Lean declaration body

```
type:
{count : ℕ} → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count → ℝ → (Fin count → ℝ) × ℝ → ENNReal

value:
fun {count : ℕ} (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (rate : ℝ)
  (p : (Fin count → ℝ) × ℝ) =>
  M.endpointWeight p * LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalNoArrivalTail rate p.2
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.endpointWeight](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalNoArrivalTail](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.rawGapTailDensity

Declaration location: paper-local

Lean declaration body

```
type:
{count : ℕ} → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count → ℝ → (Fin count → ℝ) × ℝ → ENNReal

value:
fun {count : ℕ} (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (rate : ℝ)
  (p : (Fin count → ℝ) × ℝ) =>
  AppliedModelingLib.Probability.PoissonProcess.exponentialBlockDensity rate count p.1 * M.terminalWeight rate p
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.exponentialBlockDensity](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalWeight](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.theorem2CausalConditionalLikelihood

Declaration location: paper-local

Lean declaration body

```
type:
(T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) →
  LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel T.count → ℝ → ℝ

value:
fun (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline)
  (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel T.count) (rate : ℝ) =>
  (M.rawGapTailDensity rate (T.gap, T.tail)).toReal
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.gap](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.tail](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.rawGapTailDensity](#)

LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood

Declaration location: paper-local

Lean declaration body

```
type:
(T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) →
  (M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count) →
  M.EndpointDensityPresentation → ℝ → ℝ

value:
fun (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline)
  (M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count)
  (D : M.EndpointDensityPresentation) (rate : ℝ) =>
  LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.theorem2CausalConditionalLikelihood T
  (LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation.toDensityModel M
  D)
  rate
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation.toDensityModel](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.theorem2CausalConditionalLikelihood](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.toFiniteStageCausalEndpointProfile

Declaration location: paper-local

Lean declaration body

```
type:
  {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel T.count →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T

value:
  fun {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
    (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel T.count) =>
  { startDensity := M.startWeight.toReal,
    startDensity_nonnegative :=
      LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.toFiniteStageCausalEndpointProfile._proof_1 M,
    endKernel := M.endKernel,
    endKernel_isMarkov :=
      LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.toFiniteStageCausalEndpointProfile._proof_2 M,
    endDensity := M.endDensity,
    endDensity_measurable :=
      LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.toFiniteStageCausalEndpointProfile._proof_3 M,
    endKernel_eq_withDensity :=
      LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.toFiniteStageCausalEndpointProfile._proof_5 M }
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile](#)

LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.observedPrefix

Declaration location: paper-local

Lean declaration body

```
type:
  (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) → (j : Fin (T.count + 1)) → Fin ↑j → ℝ

value:
  fun (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) (j : Fin (T.count + 1))
    (a : Fin ↑j) =>
    LBG24SpatialUnderreporting.finiteArrivalPrefix T.gap j a
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.gap](#)
- [LBG24SpatialUnderreporting.finiteArrivalPrefix](#)

LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.stageSurvival

Declaration location: paper-local

Lean declaration body

```
type:
  {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T → Fin T.count → ℝ

value:
  fun {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
    (M : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) (i : Fin T.count) =>
    (((M.endKernel i.castSucc) (LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.observedPrefix T i.castSucc))
      (Set.loi (T.gap i))).toReal
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.gap](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.observedPrefix](#)

LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.terminalEndpointDensity

Declaration location: paper-local

Lean declaration body

```
type:
  {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T → ℝ

value:
  fun {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
    (M : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) =>
    (M.endDensity (Fin.last T.count)
     (LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.observedPrefix T (Fin.last T.count)) T.tail).toReal
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.tail](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.observedPrefix](#)

LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.endpointWeight

Declaration location: paper-local

Lean declaration body

```
type:
  {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T → ℝ

value:
  fun {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
    (M : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) =>
    M.startDensity * M.terminalEndpointDensity * ∏ i : Fin T.count, M.stageSurvival i
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.stageSurvival](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.terminalEndpointDensity](#)

LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.correctedResidual

Declaration location: paper-local

Lean declaration body

```
type:
  {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T → ℝ

value:
  fun {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
    (M : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) =>
    M.endpointWeight * (↑ T.count.factorial / T.window.exposure ^ T.count)
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow.exposure](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.endpointWeight](#)

LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual

Declaration location: paper-local

Lean declaration body

```
type:
(T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) →
(M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count) → M.EndpointDensityPresentation → ℝ

value:
fun (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline)
  (M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count)
  (D : M.EndpointDensityPresentation) =>
  (LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation.toDensityModel M
   D).toFiniteStageCausalEndpointProfile.correctedResidual
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation.toDensityModel](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.toFiniteStageCausalEndpointProfile](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.correctedResidual](#)

LBG24SpatialUnderreporting.mleRate

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℝ

value:
fun (totalCount : ℕ) (totalExposure : ℝ) => ↑totalCount / totalExposure
```

LBG24SpatialUnderreporting.poissonRateLogLikelihood

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℝ → ℝ

value:
fun (totalCount : ℕ) (totalExposure rate : ℝ) =>
  AppliedModelingLib.Probability.PoissonProcess.poissonRateLogLikelihoodKernel totalCount totalExposure rate
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.poissonRateLogLikelihoodKernel](#)

LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℕ → ℝ

value:
fun (rate exposure : ℝ) (count : ℕ) => AppliedModelingLib.Probability.PoissonProcess.countLikelihood rate exposure count
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.countLikelihood](#)

LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood

Declaration location: paper-local

Lean declaration body

type:

$\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$

value:

fun (gamma rate exposure : $\mathbb{R}$) (count : $\mathbb{N}$) =>

AppliedModelingLib.Probability.PoissonProcess.zeroInflatedCountLikelihood gamma rate exposure count

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.zeroInflatedCountLikelihood](#)

Material library prerequisites

AppliedModelingLib.Probability.PoissonProcess.rateExposureParam

Source locator: cited publication, lines 30-41

Verbatim paper-source connection

$$p(\tilde{M}_i; \lambda(e_i - s_i)) = \frac{(\lambda(e_i - s_i))^{\tilde{M}_i}}{\tilde{M}_i!} \exp(-\lambda(e_i - s_i))$$

(2)

[form-feed in source extraction]
is the probability mass function of a Poisson distributed random variable with $\tilde{M}_i$ occurrences and rate parameter $\lambda(e_i - s_i)$, and $f(D_i)$ is a function that does not depend on λ .

[Next verbatim source excerpt]

As reference, the likelihood of a Poisson distribution with $M \geq 0$ occurrences and rate $\lambda(e-s) > 0$ is as follows:
$$p(M | \lambda(e-s)) = \frac{(\lambda(e-s))^M}{M!} \exp(-\lambda(e-s)).$$

Lean-expanded library semantic target

type:
(rate exposure : ℝ) → 0 ≤ rate * exposure → NNReal

value:
fun (rate exposure : ℝ) (h_nonneg : 0 ≤ rate * exposure) => (rate * exposure, h_nonneg)

Exact Lean library declaration

```
/-- Nonnegative rate-exposure product as the `ℝ≥0` parameter of a Poisson law. -/  
def rateExposureParam (rate exposure : ℝ)  
  (h_nonneg : 0 ≤ rate * exposure) : ℝ≥0 :=  
  (rate * exposure, h_nonneg)
```

Recorded source-to-library screening Verdict: matches

Reason: It is exactly Equation (2)'s nonnegative Poisson mean lambda times (e minus s), with the required nonnegativity witness.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw

Source locator: cited publication, lines 202-204

Verbatim paper-source connection

Model part 2: reports. An incident i of type θ is active in time window $[t_i, t_i + T_i]$. During this time, reports are submitted according to a homogeneous Poisson process, with rate $\lambda_\theta > 0$. Thus, the number of reports M_i for incident i follows a Poisson distribution, $M_i \sim \text{Poisson}(\lambda_\theta T_i)$.

[Next verbatim source excerpt]

Theorem 2. Consider a Poisson process $\{X(t), T \geq t \geq 0\}$ with rate λ (where $X(t)$ denotes the number of jumps up to time t), and denote the jumping times $\{T_1, T_2, \dots\}$. Denote the random

Lean declaration type (metadata)

```
field isProbability:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P),
  MeasureTheory.IsProbabilityMeasure P

field rate:
  {Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P → ℝ

field rate_pos:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P),
  0 < self.rate

field count:
  {Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P → NNReal → Ω → ℝ

field count_measurable:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P) (t : NNReal),
  Measurable (self.count t)

field count_zero_ae:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P),
  ∀m (ω : Ω) ∂P, self.count 0 ω = 0

field count_mono_ae:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P),
  ∀m (ω : Ω) ∂P, Monotone fun (t : NNReal) => self.count t ω

field hasIndepIncements:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P),
  ProbabilityTheory.HasIndepIncements self.count P

field increment_hasLaw:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P)
  {s t : NNReal} (hst : s ≤ t),
  ProbabilityTheory.HasLaw (fun (ω : Ω) => self.count t ω - self.count s ω)
  (ProbabilityTheory.poissonMeasure
    (AppliedModelingLib.Probability.PoissonProcess.rateExposureParam self.rate (↑t - ↑s)
      (mul_nonneg (le_of_lt self.rate_pos) (sub_nonneg.mpr (NNReal.coe_le_coe.mpr hst))))))
  P
```

Exact Lean library declaration

/--
A positive-rate homogeneous Poisson counting process on a forward time axis.

The probability-law field deliberately makes this suitable for a stationary/Palm law once the caller separately constructs that law. It says nothing about how that Palm law was obtained. `HasIndepIncements` is the mathlib process predicate, specialized here to $\mathbb{R}_{\geq 0}$ time.

```
structure ForwardHomogeneousPoissonCountingProcessByLaw
  (Ω : Type*) [MeasurableSpace Ω] (P : Measure Ω) where
  /- The forward process is carried by a probability law. -/
  isProbability : IsProbabilityMeasure P
  rate : ℝ
  rate_pos : 0 < rate
  count : ℝ≥0 → Ω → ℕ
  count_measurable : ∀ t, Measurable (count t)
  count_zero_ae : ∀m ω ∂P, count 0 ω = 0
  count_mono_ae : ∀m ω ∂P, Monotone fun t => count t ω
  hasIndepIncements : ProbabilityTheory.HasIndepIncements count P
  increment_hasLaw :
```

```

 $\forall \{s\ t : \mathbb{R} \geq 0\} \ (hst : s \leq t),$ 
ProbabilityTheory.HasLaw
(fun  $\omega : \Omega \Rightarrow$  count  $t\ \omega$  - count  $s\ \omega$ )
(ProbabilityTheory.poissonMeasure
(rateExposureParam rate (( $t : \mathbb{R}$ ) - ( $s : \mathbb{R}$ ))
(mul_nonneg (le_of_lt rate_pos)
(sub_nonneg.mpr (NNReal.coe_le_coe.mpr hst)))))) P

```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.rateExposureParam](#)

Recorded source-to-library screening Verdict: matches

Reason: The exact structure requires a probability law, a strictly positive rate, measurable monotone counts starting at zero a.e., independent increments, and for every deterministic $s \leq t$ a Poisson increment law with parameter $\text{rate} * (t - s)$, as made exact by `rateExposureParam`. This matches the positive-rate homogeneous-Poisson reporting premise. Its displayed claim use assumes $0 < \text{observedRate}$ and asks only for processes whose count fields equal the displayed calendar counts and whose rates equal `observedRate`; it does not assert a Poisson count-law transfer at an arbitrary process-dependent endpoint.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-specific semantic prerequisites

LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel

Paper-local declaration:

LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel

Source locator: cited publication, lines 1574-1592

Verbatim paper-source connection

Condition 2: Given the jumping times of the Poisson process up to any time t , that is, for some $m \in \mathbb{Z}^+$, given $\{T_1 = t_1, \dots, T_m = t_m, T_{m+1} > t\}$, the distribution of E is $P[E = t | \{T_1 = t_1, \dots, T_m = t_m, T_{m+1} > t\}] = h_m(t)$, for some $m \in \mathbb{Z}^+$, $\forall T \geq t \geq t_m$, where $h_m(t)$ is a function independent of λ . Furthermore, given $T_1 = t_1$, the distribution of E is independent of the realization of S : $P[E = t | T_1 = t_1, S = s] = P[E = t | T_1 = t_1]$, $\forall T \geq t \geq t_1, \forall s \geq t_1$.
Now, suppose we observe data which contains for each incident, $S = s, E = e, X(e) - X(s) = M$, where $X(s) = m \geq 1$, and all the jumping times within interval $(s, e]$: $JM = \{T_{m+1} = t_{m+1}, \dots, T_{m+M} = t_{m+M}\}$, where $t_{m+1} > s$ and $T_{m+M} \leq e$. Then, conditional on the observed jumping times before s : $H := \{T_1 = t_1, \dots, T_m = t_m \leq s\}$, the likelihood of the data can be decomposed as follows:
 $P[X(e) - X(s) = M, E = e, S = s, JM | H] = f(e, s, H, JM) p(M | \lambda(e - s))$,

(8)

where $p(M | \lambda(e - s))$ denotes the likelihood of a Poisson distribution with M occurrences and rate parameter $\lambda(e - s)$, and $f(e, s, H, JM)$ is a function that does not depend on λ .
The proof of the theorem relies on the following lemma, which establishes the distribution of the (random) time between S and the first jump time after S .

Lean-elaborated paper signature

field selectedStartLikelihood:

{count : $\mathbb{N}$ } → LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count → ENNReal

field endpointResponse:

{count : $\mathbb{N}$ } →
LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count →
(j : Fin (count + 1)) → ProbabilityTheory.Kernel (Fin $\uparrow$ j → $\mathbb{R}$) $\mathbb{R}$

field endpointResponse_isMarkov:

$\forall \{count : \mathbb{N}\} (self : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count) (j : Fin (count + 1)),$
ProbabilityTheory.IsMarkovKernel (self.endpointResponse j)

field endpointResponse_nonnegative_support:

$\forall \{count : \mathbb{N}\} (self : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count) (j : Fin (count + 1))$
(history : Fin $\uparrow$ j → $\mathbb{R}$), ((self.endpointResponse j) history) (Set.lto 0) = 0

field responseSurvival_measurable:

$\forall \{count : \mathbb{N}\} (self : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count) (i : Fin count),$
Measurable fun (gaps : Fin count → $\mathbb{R}$) =>
((self.endpointResponse i.castSucc) (LBG24SpatialUnderreporting.finiteArrivalPrefix gaps i.castSucc))
(Set.lto (gaps i))

Direct retained dependencies

- [LBG24SpatialUnderreporting.finiteArrivalPrefix](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Condition 2 specifies a rate-independent endpoint law conditional on the visible jump prefix and independence from the selected start; the approved causal-stopping context identifies this as one visible-prefix, future-gap-blind endpoint kernel with the regularity and nonnegative remaining-time semantics in the target.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation

Source locator: cited publication, lines 1574-1592

Verbatim paper-source connection

Condition 2: Given the jumping times of the Poisson process up to any time t , that is, for some $m \in \mathbb{Z}^+$, given $\{T_1 = t_1, \dots, T_m = t_m, T_{m+1} > t\}$, the distribution of E is $P[E = t | \{T_1 = t_1, \dots, T_m = t_m, T_{m+1} > t\}] = h_m(t)$, for some $m \in \mathbb{Z}^+$, $\forall T \geq t \geq t_m$, where $h_m(t)$ is a function independent of λ . Furthermore, given $T_1 = t_1$, the distribution of E is independent of the realization of S : $P[E = t | T_1 = t_1, S = s] = P[E = t | T_1 = t_1]$, $\forall T \geq t \geq t_1, \forall s \geq t_1$.
Now, suppose we observe data which contains for each incident, $S = s, E = e, X(e) - X(s) = M$, where $X(s) = m \geq 1$, and all the jumping times within interval $(s, e]$: $JM = \{T_{m+1} = tm+1, \dots, T_{m+M} = tm+M\}$, where $tm+1 > s$ and $T_{m+M} \leq e$. Then, conditional on the observed jumping times before s : $H := \{T_1 = t_1, \dots, T_m = t_m \leq s\}$, the likelihood of the data can be decomposed as follows:
 $P[X(e) - X(s) = M, E = e, S = s, JM | H] = f(e, s, H, JM)p(M | \lambda(e - s))$,

(8)

where $p(M | \lambda(e - s))$ denotes the likelihood of a Poisson distribution with M occurrences and rate parameter $\lambda(e - s)$, and $f(e, s, H, JM)$ is a function that does not depend on λ .
The proof of the theorem relies on the following lemma, which establishes the distribution of the (random) time between S and the first jump time after S .

Lean-elaborated paper signature

```
field endpointDensity:
{count : ℕ} →
{M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count} →
M.EndpointDensityPresentation → (j : Fin (count + 1)) → (Fin ↑ j → ℝ) → ℝ → ENNReal

field endpointDensity_masurable:
∀ {count : ℕ} {M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count}
(self : M.EndpointDensityPresentation) (j : Fin (count + 1)), Measurable (Function.uncurry (self.endpointDensity j))

field endpointResponse_eq_withDensity:
∀ {count : ℕ} {M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count}
(self : M.EndpointDensityPresentation) (j : Fin (count + 1)),
M.endpointResponse j =
(ProbabilityTheory.Kernel.const (Fin ↑ j → ℝ) MeasureTheory.volume).withDensity (self.endpointDensity j)

field terminalDensity_masurable:
∀ {count : ℕ} {M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count}
(self : M.EndpointDensityPresentation),
Measurable fun (p : (Fin count → ℝ) × ℝ) =>
self.endpointDensity (Fin.last count) (LBG24SpatialUnderreporting.finiteArrivalPrefix p.1 (Fin.last count)) p.2
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
- [LBG24SpatialUnderreporting.finiteArrivalPrefix](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source writes the conditional endpoint term $h_m(t)$; the approved endpoint-reference-measure context explicitly treats that displayed point-density notation as the absolutely-continuous density presentation of the endpoint kernel, which is exactly the target's density/measurability branch.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.CalendarFirstReportSourceModel

Paper-local declaration:

LBG24SpatialUnderreporting.CalendarFirstReportSourceModel

Source locator: cited publication, lines 126-193

Verbatim paper-source connection

Lemma 1. Suppose each incident gets reported independently, and the distribution of the interval of reporting duration T_i has density function $f(\cdot) : [0, \infty) \rightarrow \mathbb{R}$. Then under steady state, $N_{\theta \text{observed}}$ follows a Poisson process with rate Λ_{θ} , where

$$\Lambda_{\theta} = \Lambda_{\theta} \int_0^{\infty} \exp\left(-\int_0^t \lambda_{\theta}(u) du\right) f(t) dt.$$

0

In the simplest time-homogeneous case, this rate simplifies to:

$$\Lambda_{\theta} = \Lambda_{\theta} \int_0^{\infty} \exp(-\lambda_{\theta} t) f(t) dt.$$

Proof of Lemma 1. Let $m(t)$ be the number of times an incident is reported in an interval of t , starting from its birth. We know from the model assumption that $m(t)$ follows a Poisson distribution:

$$m(t) \sim \text{Poisson} \left(\int_0^t \lambda_{\theta}(u) du \right).$$

23

In steady state, each unique incident gets reported with probability

$$\Pr[m(t) \geq 1 | T_i = t] f(t) dt = \int_0^{\infty} \left(1 - \exp\left(-\int_0^t \lambda_{\theta}(u) du\right) \right) f(t) dt,$$

which simplifies under time-homogeneity to

$$\int_0^{\infty} \exp(-\lambda_{\theta} t) f(t) dt.$$

Over a time interval of t , the total number of incidents of type θ that happen, $N_{\theta}(t)$ follows a Poisson process with rate Λ_{θ} . Conditional on $N_{\theta}(t) = n$, $n = 0, 1, \dots$, under steady state, $N_{\theta \text{observed}}(t)$ follows a binomial distribution with parameters (n, p) . Thus $N_{\theta \text{observed}}(t)$ follows a Poisson distribution with rate $\Lambda_{\theta} p$, which completes the proof.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Lean-elaborated paper signature

field incidentRate:
{ $\Omega : \text{Type } u_1$ } $\rightarrow$
[inst : MeasurableSpace Ω] $\rightarrow$
{ $P : \text{MeasureTheory.Measure } \Omega$ } $\rightarrow \text{LBG24SpatialUnderreporting.CalendarFirstReportSourceModel } \Omega P \rightarrow \mathbb{R}$

field incidentRate_nonnegative:
 $\forall \{ \Omega : \text{Type } u_1 \} \text{ [inst : MeasurableSpace } \Omega \] \{ P : \text{MeasureTheory.Measure } \Omega \}$
(self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel ΩP), $0 \leq \text{self.incidentRate}$

```

field retentionProbability:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    {P : MeasureTheory.Measure Ω} → LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P → ℝ

field retentionProbability_nonnegative:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P), 0 ≤ self.retentionProbability

field retentionProbability_le_one:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P), self.retentionProbability ≤ 1

field retainedDelay:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    {P : MeasureTheory.Measure Ω} →
      LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P → MeasureTheory.Measure ℝ

field retainedDelay_finite:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P),
    MeasureTheory.IsFiniteMeasure self.retainedDelay

field retainedDelay_nonnegative_support:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P), self.retainedDelay (Set.lio 0) = 0

field retainedDelay_mass:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P),
    self.retainedDelay Set.univ = ENNReal.ofReal self.retentionProbability

field markedBirths:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    {P : MeasureTheory.Measure Ω} →
      (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P) →
        AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω (ℝ × ℝ) P
        (ENNReal.ofReal self.incidentRate • MeasureTheory.volume.prod self.retainedDelay)

```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonCountingMeasureByLaw](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Lemma 1 gives Poisson incident arrivals, independent reporting, and a first-report-before-duration probability; the approved calendar-first-report context specifically authorizes the stationary marked birth process, finite retained-delay measure, and calendar-time first-report interpretation. The target correctly allows zero incidence and zero retention.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.iidGapLaw

Source locator: cited publication, lines 202-204

Verbatim paper-source connection

Model part 2: reports. An incident i of type θ is active in time window $[t_i, t_i + T_i]$. During this time, reports are submitted according to a homogeneous Poisson process, with rate $\lambda_\theta > 0$. Thus, the number of reports M_i for incident i follows a Poisson distribution, $M_i \sim \text{Poisson}(\lambda_\theta T_i)$.

[Next verbatim source excerpt]

Theorem 2. Consider a Poisson process $\{X(t), T \geq t \geq 0\}$ with rate λ (where $X(t)$ denotes the number of jumps up to time t), and denote the jumping times $\{T_1, T_2, \dots\}$. Denote the random

Lean-expanded paper semantic target

type:
(count : ℕ) → ℝ → MeasureTheory.Measure (Fin count → ℝ)

value:
fun (count : ℕ) (rate : ℝ) => MeasureTheory.Measure.pi fun (x : Fin count) => ProbabilityTheory.expMeasure rate

Recorded source-to-declaration screening Verdict: matches

Reason: The source requires positive-rate homogeneous-Poisson reporting and uses finite iid exponential interarrival gaps. The exact Lean value is the product measure on Fin count with every coordinate distributed as expMeasure rate, so at every source-admissible positive rate it is exactly a finite iid exponential-gap block; accepting additional rate arguments does not remove or alter any source case.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition

Paper-local declaration:

LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition

Source locator: cited publication, lines 126-193

Verbatim paper-source connection

Lemma 1. Suppose each incident gets reported independently, and the distribution of the interval of reporting duration T_i has density function $f(\cdot) : [0, \infty) \rightarrow \mathbb{R}$. Then under steady state, $N_{\theta}^{\text{observed}}$ follows a Poisson process with rate Λ_{θ} , where

$$\Lambda_{\theta} = \Lambda_{\theta} \int_0^{\infty} \exp\left(-\int_0^t \lambda_{\theta}(u) du\right) f(t) dt.$$

0

In the simplest time-homogeneous case, this rate simplifies to:

$$\Lambda_{\theta} = \Lambda_{\theta} \int_0^{\infty} \exp(-\lambda_{\theta} t) f(t) dt.$$

Proof of Lemma 1. Let $m(t)$ be the number of times an incident is reported in an interval of t , starting from its birth. We know from the model assumption that $m(t)$ follows a Poisson distribution:

$$m(t) \sim \text{Poisson} \left(\int_0^t \lambda_{\theta}(u) du \right).$$

23

In steady state, each unique incident gets reported with probability

$$\Pr[m(t) \geq 1 | T_i = t] f(t) dt = \int_0^{\infty} \left(1 - \exp\left(-\int_0^t \lambda_{\theta}(u) du\right) \right) f(t) dt,$$

which simplifies under time-homogeneity to

$$\int_0^{\infty} \exp(-\lambda_{\theta} t) f(t) dt.$$

Over a time interval of t , the total number of incidents of type θ that happen, $N_{\theta}(t)$ follows a Poisson process with rate Λ_{θ} . Conditional on $N_{\theta}(t) = n$, $n = 0, 1, \dots$, under steady state, $N_{\theta}^{\text{observed}}(t)$ follows a binomial distribution with parameters (n, p) . Thus $N_{\theta}^{\text{observed}}(t)$ follows a Poisson distribution with rate $\Lambda_{\theta} p$, which completes the proof.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Lean-elaborated paper signature

field integrable:

$\forall \{ \text{durationDensity} : \mathbb{R} \rightarrow \mathbb{R} \},$
 LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition durationDensity $\rightarrow$
 MeasureTheory.Integrable durationDensity (MeasureTheory.volume.restrict (Set.Ici 0))

field nonnegative_ae:

$\forall \{ \text{durationDensity} : \mathbb{R} \rightarrow \mathbb{R} \},$
 LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition durationDensity $\rightarrow$
 $\forall^m (t : \mathbb{R}) \partial \text{MeasureTheory.volume.restrict (Set.Ici 0)}, 0 \leq \text{durationDensity } t$

field integral_eq_one:
 $\forall \{ \text{durationDensity} : \mathbb{R} \rightarrow \mathbb{R} \},$
 LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition durationDensity $\rightarrow$
 $\int (t : \mathbb{R}) \text{ in Set.Ici } 0, \text{ durationDensity } t = 1$

Recorded source-to-declaration screening Verdict: matches

Reason: The pinned Lemma 1 text calls f a density on $[0, \text{infinity})$ and integrates it as the duration law; nonnegativity, unit mass, and integrability on that source domain are the exact density semantics represented by the target.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel

Paper-local declaration:

LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel

Source locator: cited publication, lines 194-201

Verbatim paper-source connection

Model part 1: incident birth and death. Incidents of type θ occur (“are born”) according to a random process parameterized by Λ_θ . (For example, a homogeneous Poisson process with rate Λ_θ). Type θ encodes geographic and incident-level (e.g., whether it is a hazard) characteristics. For each incident i , let t_i denote its birth time. Furthermore, the incident “dies” at some time $t_i + T_i$, and no more reports are logged. The death time reflects, for example, the incident being resolved by the agency or by an outside group, or reports no longer being logged because the issue has been inspected. Crucially, birth times are unobserved in the model; death times may be unobserved or partially observed.⁴

Lean-elaborated paper signature

field birthTime:

{Incident : Type u_1} →
{Report : Type u_2} → LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel Incident Report → Incident → ℝ

field duration:

{Incident : Type u_1} →
{Report : Type u_2} → LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel Incident Report → Incident → ℝ

field duration_nonnegative:

∀ {Incident : Type u_1} {Report : Type u_2}
(self : LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel Incident Report) (incident : Incident),
0 ≤ self.duration incident

field reportIncident:

{Incident : Type u_1} →
{Report : Type u_2} → LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel Incident Report → Report → Incident

field reportTime:

{Incident : Type u_1} →
{Report : Type u_2} → LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel Incident Report → Report → ℝ

field report_during_active_window:

∀ {Incident : Type u_1} {Report : Type u_2}
(self : LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel Incident Report) (report : Report),
self.birthTime (self.reportIncident report) ≤ self.reportTime report ∧
self.reportTime report ≤ self.birthTime (self.reportIncident report) + self.duration (self.reportIncident report)

Recorded source-to-declaration screening Verdict: matches

Reason: The source makes t_i the incident birth time, $t_i + T_i$ its later death time, and says that no reports are logged after death. The exact constructor is the corresponding incident/report carrier: T_i is represented by a nonnegative duration, and every report linked to an incident is temporally between that incident’s birth and death.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition

Paper-local declaration:

LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition

Source locator: cited publication, lines 126-193

Verbatim paper-source connection

Lemma 1. Suppose each incident gets reported independently, and the distribution of the interval of reporting duration T_i has density function $f(\cdot) : [0, \infty) \rightarrow \mathbb{R}$. Then under steady state, $N_{\theta}^{\text{observed}}$ follows a Poisson process with rate Λ_{θ} , where

$$\Lambda_{\theta} = \Lambda_{\theta} \int_0^{\infty} \exp\left(-\int_0^t \lambda_{\theta}(u) du\right) f(t) dt.$$

0

In the simplest time-homogeneous case, this rate simplifies to:

$$\Lambda_{\theta} = \Lambda_{\theta} \int_0^{\infty} \exp(-\lambda_{\theta} t) f(t) dt.$$

Proof of Lemma 1. Let $m(t)$ be the number of times an incident is reported in an interval of t , starting from its birth. We know from the model assumption that $m(t)$ follows a Poisson distribution:

$$m(t) \sim \text{Poisson} \left(\int_0^t \lambda_{\theta}(u) du \right).$$

23

In steady state, each unique incident gets reported with probability

$$\Pr[m(t) \geq 1 | T_i = t] f(t) dt = \int_0^{\infty} \left(1 - \exp\left(-\int_0^t \lambda_{\theta}(u) du\right) \right) f(t) dt,$$

which simplifies under time-homogeneity to

$$\int_0^{\infty} \exp(-\lambda_{\theta} t) f(t) dt.$$

Over a time interval of t , the total number of incidents of type θ that happen, $N_{\theta}(t)$ follows a Poisson process with rate Λ_{θ} . Conditional on $N_{\theta}(t) = n$, $n = 0, 1, \dots$, under steady state, $N_{\theta}^{\text{observed}}(t)$ follows a binomial distribution with parameters (n, p) . Thus $N_{\theta}^{\text{observed}}(t)$ follows a Poisson distribution with rate $\Lambda_{\theta} p$, which completes the proof.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Lean-elaborated paper signature

field nonnegative:

$\forall \{\text{reportingIntensity} : \mathbb{R} \rightarrow \mathbb{R}\},$
LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition reportingIntensity $\rightarrow$
 $\forall (t : \mathbb{R}), 0 \leq t \rightarrow 0 \leq \text{reportingIntensity } t$

field intervalIntegrable:

$\forall \{\text{reportingIntensity} : \mathbb{R} \rightarrow \mathbb{R}\},$
LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition reportingIntensity $\rightarrow$
 $\forall (t : \mathbb{R}), 0 \leq t \rightarrow \text{IntervalIntegrable } \text{reportingIntensity } \text{MeasureTheory.volume } 0 \ t$

Recorded source-to-declaration screening Verdict: matches

Reason: Lemma 1's age-varying report count has Poisson mean $\int_0^t \lambda(u) du$ for every nonnegative duration; treating λ as a reporting intensity gives the target's nonnegativity and finite-interval integrability without adding strict positivity.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

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LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel

Paper-local declaration:

LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel

Source locator: cited publication, lines 126-193

Verbatim paper-source connection

Lemma 1. Suppose each incident gets reported independently, and the distribution of the interval of reporting duration T_i has density function $f(\cdot) : [0, \infty) \rightarrow \mathbb{R}$. Then under steady state, $N_{\theta}^{\text{observed}}$ follows a Poisson process with rate Λ_{θ} , where

$$\Lambda_{\theta} = \Lambda_{\theta} \int_0^{\infty} \exp\left(-\int_0^t \lambda_{\theta}(u) du\right) f(t) dt.$$

0

In the simplest time-homogeneous case, this rate simplifies to:

$$\Lambda_{\theta} = \Lambda_{\theta} \int_0^{\infty} \exp(-\lambda_{\theta} t) f(t) dt.$$

Proof of Lemma 1. Let $m(t)$ be the number of times an incident is reported in an interval of t , starting from its birth. We know from the model assumption that $m(t)$ follows a Poisson distribution:

$$m(t) \sim \text{Poisson}\left(\int_0^t \lambda_{\theta}(u) du\right).$$

23

In steady state, each unique incident gets reported with probability

$$\Pr[m(t) \geq 1 | T_i = t] f(t) dt = \int_0^{\infty} \left(1 - \exp\left(-\int_0^t \lambda_{\theta}(u) du\right)\right) f(t) dt,$$

which simplifies under time-homogeneity to

$$\int_0^{\infty} \exp(-\lambda_{\theta} t) f(t) dt.$$

Over a time interval of t , the total number of incidents of type θ that happen, $N_{\theta}(t)$ follows a Poisson process with rate Λ_{θ} . Conditional on $N_{\theta}(t) = n$, $n = 0, 1, \dots$, under steady state, $N_{\theta}^{\text{observed}}(t)$ follows a binomial distribution with parameters (n, p) . Thus $N_{\theta}^{\text{observed}}(t)$ follows a Poisson distribution with rate $\Lambda_{\theta} p$, which completes the proof.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Lean-elaborated paper signature

field reportingIntensity:

{ Ω : Type u_1} →
[inst : MeasurableSpace Ω] →
{P : MeasureTheory.Measure Ω } → LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P → $\mathbb{R} \rightarrow \mathbb{R}$

field reportingIntensity_source:

∀ { Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω }
{self : LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P},
LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition self.reportingIntensity

```

field durationDensity:
{Ω : Type u_1} →
[inst : MeasurableSpace Ω] →
{P : MeasureTheory.Measure Ω} → LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P → ℝ → ℝ

field durationDensity_source:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
(self : LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P),
LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition self.durationDensity

field calendarFirstReports:
{Ω : Type u_1} →
[inst : MeasurableSpace Ω] →
{P : MeasureTheory.Measure Ω} →
LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P →
LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P

field preFirstReport_survival_retention_probability:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
(self : LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P),
self.calendarFirstReports.retentionProbability =
LBG24SpatialUnderreporting.continuousDurationFirstReportProbabilityOfCumulativeIntensity
(fun (t : ℝ) => ∫ (u : ℝ) in 0..t, self.reportingIntensity u) self.durationDensity

```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel](#)
- [LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition](#)
- [LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition](#)
- [LBG24SpatialUnderreporting.continuousDurationFirstReportProbabilityOfCumulativeIntensity](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Under the approved calendar-first-report reading, the stationary marked Poisson model and retained first-report mass are exactly the Lemma 1 reporting-before-duration probability.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel

Paper-local declaration:

LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel

Source locator: cited publication, lines 126-193

Verbatim paper-source connection

Lemma 1. Suppose each incident gets reported independently, and the distribution of the interval of reporting duration T_i has density function $f(\cdot) : [0, \infty) \rightarrow \mathbb{R}$. Then under steady state, $N_{\theta}^{\text{observed}}$ follows a Poisson process with rate Λ_{θ} , where

$$\Lambda_{\theta} = \Lambda_{\theta} \int_0^{\infty} \exp\left(-\int_0^t \lambda_{\theta}(u) du\right) f(t) dt.$$

0

In the simplest time-homogeneous case, this rate simplifies to:

$$\Lambda_{\theta} = \Lambda_{\theta} \int_0^{\infty} \exp(-\lambda_{\theta} t) f(t) dt.$$

Proof of Lemma 1. Let $m(t)$ be the number of times an incident is reported in an interval of t , starting from its birth. We know from the model assumption that $m(t)$ follows a Poisson distribution:

$$m(t) \sim \text{Poisson} \left(\int_0^t \lambda_{\theta}(u) du \right).$$

23

In steady state, each unique incident gets reported with probability

$$\Pr[m(t) \geq 1 | T_i = t] f(t) dt = \int_0^{\infty} \left(1 - \exp\left(-\int_0^t \lambda_{\theta}(u) du\right) \right) f(t) dt,$$

which simplifies under time-homogeneity to

$$\int_0^{\infty} \exp(-\lambda_{\theta} t) f(t) dt.$$

Over a time interval of t , the total number of incidents of type θ that happen, $N_{\theta}(t)$ follows a Poisson process with rate Λ_{θ} . Conditional on $N_{\theta}(t) = n$, $n = 0, 1, \dots$, under steady state, $N_{\theta}^{\text{observed}}(t)$ follows a binomial distribution with parameters (n, p) . Thus $N_{\theta}^{\text{observed}}(t)$ follows a Poisson distribution with rate $\Lambda_{\theta} p$, which completes the proof.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Lean-elaborated paper signature

field reportingRate:

$\{\Omega : \text{Type } u_1\} \rightarrow$
 $[\text{inst} : \text{MeasurableSpace } \Omega] \rightarrow$
 $\{P : \text{MeasureTheory.Measure } \Omega\} \rightarrow \text{LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel } \Omega \ P \rightarrow \mathbb{R}$

field reportingRate_nonnegative:

$\forall \{\Omega : \text{Type } u_1\} [\text{inst} : \text{MeasurableSpace } \Omega] \{P : \text{MeasureTheory.Measure } \Omega\}$
 $(\text{self} : \text{LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel } \Omega \ P), 0 \leq \text{self.reportingRate}$


```

field durationDensity:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    {P : MeasureTheory.Measure Ω} → LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Ω P → ℝ → ℝ

field durationDensity_source:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Ω P),
  LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition self.durationDensity

field calendarFirstReports:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    {P : MeasureTheory.Measure Ω} →
      LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Ω P →
        LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P

field preFirstReport_survival_retention_probability:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Ω P),
  self.calendarFirstReports.retentionProbability =
    LBG24SpatialUnderreporting.continuousDurationFirstReportProbability self.reportingRate self.durationDensity

```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel](#)
- [LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition](#)
- [LBG24SpatialUnderreporting.continuousDurationFirstReportProbability](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The homogeneous specialization identifies the retained first-report probability with the source's displayed 1 - integral $\exp(-\lambda t) f(t) dt$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.Theorem2ConditionOneSelection

Paper-local declaration:

LBG24SpatialUnderreporting.Theorem2ConditionOneSelection

Source locator: cited publication, lines 1571-1573

Verbatim paper-source connection

Condition 1: Given $T_1 = t_1$, S is defined on $[t_1, T]$, and $P(S = t | T_1 = t_1) = g(t)$, $\forall t \geq t_1$, where $g(t)$ is not dependent on λ or the sample path of the Poisson process after t_1 .

Lean-elaborated paper signature

```
field firstReportTime:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  [inst_1 : StandardBorelSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  {Tail : Type u_2} →
  [inst_3 : MeasurableSpace Tail] →
  LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail → Ω → NNReal

field startTime:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  [inst_1 : StandardBorelSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  {Tail : Type u_2} →
  [inst_3 : MeasurableSpace Tail] →
  LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail → Ω → NNReal

field observationHorizon:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  [inst_1 : StandardBorelSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  {Tail : Type u_2} →
  [inst_3 : MeasurableSpace Tail] → LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail → NNReal

field postFirstReportTail:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  [inst_1 : StandardBorelSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  {Tail : Type u_2} →
  [inst_3 : MeasurableSpace Tail] →
  LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail → Ω → Tail

field rateFreeStartKernel:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  [inst_1 : StandardBorelSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  {Tail : Type u_2} →
  [inst_3 : MeasurableSpace Tail] →
  LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail → ProbabilityTheory.Kernel NNReal NNReal

field rateFreeStartKernel_isMarkov:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail),
  ProbabilityTheory.IsMarkovKernel self.rateFreeStartKernel

field firstReportTime_measurable:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail), Measurable self.firstReportTime

field startTime_measurable:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail), Measurable self.startTime

field postFirstReportTail_measurable:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail), Measurable self.postFirstReportTail

field firstReport_le_start:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail) (ω : Ω),
  self.firstReportTime ω ≤ self.startTime ω

field start_le_observationHorizon:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
```

```

[inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
(self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection  $\bar{\Omega}$  P Tail) ( $\omega$  :  $\Omega$ ),
self.startTime  $\omega \leq$  self.observationHorizon

field start_conditional_law:
 $\forall$  { $\Omega$  : Type u_1} [inst : MeasurableSpace  $\Omega$ ] [inst_1 : StandardBorelSpace  $\Omega$ ] {P : MeasureTheory.Measure  $\Omega$ }
[inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
(self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection  $\bar{\Omega}$  P Tail),
 $\forall^m (\omega : \Omega) \partial P,$ 
  ProbabilityTheory.HasLaw self.startTime (self.rateFreeStartKernel (self.firstReportTime  $\omega$ ))
  ((ProbabilityTheory.condExpKernel P (MeasurableSpace.comap self.firstReportTime inferInstance))  $\omega$ )

field rateFreeStartKernel_supported_after_first:
 $\forall$  { $\Omega$  : Type u_1} [inst : MeasurableSpace  $\Omega$ ] [inst_1 : StandardBorelSpace  $\Omega$ ] {P : MeasureTheory.Measure  $\Omega$ }
[inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
(self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection  $\bar{\Omega}$  P Tail) (t : NNReal),
(self.rateFreeStartKernel t) (Set.Ici t) = 1

field start_condIndep_postFirstReportTail:
 $\forall$  { $\Omega$  : Type u_1} [inst : MeasurableSpace  $\Omega$ ] [inst_1 : StandardBorelSpace  $\Omega$ ] {P : MeasureTheory.Measure  $\Omega$ }
[inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
(self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection  $\bar{\Omega}$  P Tail),
ProbabilityTheory.CondIndepFun (MeasurableSpace.comap self.firstReportTime inferInstance)
(Measurable.comap_le self.firstReportTime_measurable) self.startTime self.postFirstReportTail P

```

Recorded source-to-declaration screening Verdict: matches

Reason: Under the approved fixed-model Condition 1 reading, the kernel is supported on $[T_1, T]$, is rate-free as a model component, and is conditionally independent of the post-first-report tail.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.equation33_nyc_observation_end

Paper-local declaration:

LBG24SpatialUnderreporting.equation33_nyc_observation_end

Source locator: cited publication, lines 248-272

Verbatim paper-source connection

Constructing an observation interval $(S_i, E_i]$ As outlined in Theorem 1, we must be careful in how we choose an observation interval $(S_i, E_i]$ in which we count reports – we need that the interval is inside the incident lifetime, i.e., we must end the interval before the incident is addressed, $E_i \leq t_i + T_i$. Both endpoints of the interval must also satisfy the conditions outlined in Theorem 1. As discussed above, the best choice to start the observation period S_i is the time of the first report, but choosing the observation end is a design choice.

We make the following choice. Let t_{INSP} be the inspection time of incident i , and, t_{WO} be the

i
 i
time that a work order is placed for incident i , if applicable. Then, E_i of each incident i is:

{
 $E_i = \min 100 \text{ days} + S_i, t_{\text{INSP}}$
, t_{WO}

(33)

i
 i
The maximum duration of 100 is a design choice, for which we perform robustness checks (with 30 and 200 days); a maximum mitigates – for incidents not inspected for a long time – the inclusion of a time period in which an issue might have been resolved before an inspection, which would bias our estimates downward. Equation (33) requires that inspection and work order times are stopping times; that they do not depend on the future trivially holds, and it is likely that they do not depend on the reporting rate except through the type θ and the sample path number of reports received

Lean-expanded paper semantic target

type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$

value:
fun (startTime inspectionTime workOrderTime : $\mathbb{R}$) => min (min (startTime + 100) inspectionTime) workOrderTime

Recorded source-to-declaration screening Verdict: matches

Reason: Equation (33) ends the NYC observation interval at the minimum of 100 days after S_i , the inspection time, and the work-order time. The exact Lean value is that same three-way minimum, with only binary `min` association made explicit.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

LBG24SpatialUnderreporting.equation34_chicago_observation_end

Source locator: cited publication, lines 273-285

Verbatim paper-source connection

Constructing an observation interval (S_i , E_i] For Chicago reports, we make the following choice. Let t_i be the first time an incident is reported, t_{CLOSED} be the time an incident is marked closed, and $\tilde{t}$ be the time that we retrieved the dataset (and that it was last updated), 7/4/2022 at 21:40 EDT. Then, E_i of each incident i is:

$$E_i = \min 100 \text{ days} + t_i, t_{CLOSED}, \tilde{t}.$$

(34)

The choice of adding $\tilde{t}$ ensures that for “open” incidents, the observation interval is not overly long, which could bias our estimates for the reporting rate downward.

Lean-expanded paper semantic target

type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$

value:
 $\text{fun } (\text{firstReportTime } \text{closedTime } \text{retrievalTime} : \mathbb{R}) \Rightarrow \min (\min (\text{firstReportTime} + 100) \text{closedTime}) \text{retrievalTime}$

Recorded source-to-declaration screening Verdict: matches
Reason: Equation (34) ends the Chicago observation interval at the minimum of 100 days after the first report, the closed time, and the dataset retrieval time. The exact Lean value is the identical three-way minimum.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

LBG24SpatialUnderreporting.poissonRegressionRate

Paper-local declaration:

LBG24SpatialUnderreporting.poissonRegressionRate

Source locator: cited publication, lines 64-72

Verbatim paper-source connection

Second, we must empirically estimate reporting rates given high dimensional incident types θ . We leverage Poisson regression, i.e., we assume the form

$$\lambda\theta = \exp \alpha + \beta^T \theta .$$

To better match the data distribution, we further fit a zero-inflation component (corresponding to the probability that we receive no reports in the given period). To accommodate the high dimensionality of the incident type vector θ , we fit our models in a Bayesian setting and report posterior distributions of regression coefficients α and β .

[Next verbatim source excerpt]

First, we assume that the reporting rate is affected by the type vector θ as in a Poisson regression: for each model, we have a standard reporting rate $\lambda\theta$, defined as:

$$\lambda\theta = \exp \alpha + \beta^T \theta ,$$

and so, for each incident i , the likelihood follows from Equation (1) and satisfies the following:

$$L(\lambda\theta_i | D_i) \propto p(\tilde{M}_i ; \lambda\theta_i (e_i - s_i)) ,$$

(6)
13

[form-feed in source extraction]

where β is a coefficient vector of appropriate dimension, (α, β) are coefficients to be estimated, and $p(\cdot)$ is the Poisson likelihood function as defined in Equation (2).

Lean-expanded paper semantic target

type:
{Feature : Type u_1} → [Fintype Feature] → ℝ → (Feature → ℝ) → (Feature → ℝ) → ℝ

value:
fun {Feature : Type u_1} [Fintype Feature] (alpha : ℝ) (beta theta : Feature → ℝ) =>
Real.exp (alpha + $\sum_j$: Feature, beta j * theta j)

Recorded source-to-declaration screening Verdict: matches

Reason: Equations (4) and (5) specify $\lambda\theta = \exp(\alpha + \beta^T \theta)$. The exact Lean value expands the finite dot product $\beta^T \theta$ as a sum over the finite feature type and applies the same exponential link.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.zeroInflatedIndependentIncidentFamilyLikelihood

Paper-local declaration:

LBG24SpatialUnderreporting.zeroInflatedIndependentIncidentFamilyLikelihood

Source locator: cited publication, lines 87-93

Verbatim paper-source connection

We further fit a zero-inflated Poisson regression [19], which corresponds to a data generating process in which, independently for each incident, only one report can be generated with probability γ , and thus there are 0 duplicate reports: $M_i = 0.9$ For example, perhaps those incidents are personal requests for service (e.g., to plant a tree somewhere or to prune a tree to enable certain construction), for which duplicates are unlikely; alternatively, officials may physically mark the location of an incident as “report received; full inspection pending,” preventing future reports. Under this case, the likelihood of the data satisfies:

Lean-expanded paper semantic target

```
type:
{Incident : Type u_1} →
  Finset Incident → (Incident → LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel) → (Incident → ℕ) → ℝ

value:
fun {Incident : Type u_1} (incidents : Finset Incident)
  (model : Incident → LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel) (count : Incident → ℕ) =>
  ∏ i ∈ incidents,
    LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood (model i).gamma (model i).rate (model i).exposure
    (count i)
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel](#)
- [LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The target is the product over independent incidents of the observed structural-zero or ordinary-Poisson duplicate-count mixture likelihood.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 184-190

Verbatim source input

Figure 1: Model of resident reports of an incident i of type θ . Observed and unobserved events are marked with solid and dotted lines, respectively. In this example, there are a total of 3 reports made about this incident. When the reports are generated according to a homogeneous Poisson process with rate λ_θ , the average reporting delay is λ_θ^{-1} . Incident 'death' refers to the incident being resolved or otherwise marked such that no future reports are submitted and logged. The goal is to estimate the reporting rate λ_θ (and consequently, the unobserved reporting delay) for each incident of type θ .

[Next verbatim source excerpt]

Research question. We are interested in auditing the efficiency and equity of the reporting process: how quickly are incidents reported, and how does that reporting rate λ_θ vary with the incident type θ ? Estimating this value directly reveals heterogeneous reporting delays, as the expected time between incident occurrence and first report in the model is λ_θ^{-1} . The primary

Expanded PaperInterface specification

$\forall (\text{rate} : \mathbb{R}) (\text{rate_pos} : 0 < \text{rate}), \int (x : \mathbb{R}), x \partial \{ \text{rate} := \text{rate}, \text{rate_pos} := \text{rate_pos} \}. \text{measure} = 1 / \text{rate}$

Retained governing declarations

- [AppliedModelingLib.Probability.Exponential.Model](#)
- [AppliedModelingLib.Probability.Exponential.Model.measure](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{R}$
2. assumption: $0 < \text{rate}$
3. conclusion: $\int (x : \mathbb{R}), x \partial \{ \text{rate} := \text{rate}, \text{rate_pos} := \text{rate_pos} \}. \text{measure} = 1 / \text{rate}$

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceHomogeneousReportingDelayMean

Recorded source-to-Spec screening Verdict: matches

Reason: The source says that a homogeneous Poisson reporting process of rate λ_θ has expected first-report delay $1/\lambda_\theta$. The Lean target assumes a strictly positive real rate and states that the mean of the corresponding exponential probability measure is exactly $1/\text{rate}$, with the same parameter and no extra conclusion.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 1565-1590

Verbatim source input

The following Theorem 2 is a restated version of Theorem 1, where we use standard Poisson process notation, and state mathematically the two conditions we introduce in Theorem 1. Theorem 2. Consider a Poisson process $\{X(t), T \geq t \geq 0\}$ with rate λ (where $X(t)$ denotes the number of jumps up to time t), and denote the jumping times $\{T_1, T_2, \dots\}$. Denote the random variables of the start and end of an observation period S and E , and suppose S and E satisfy the following conditions, respectively:

Condition 1: Given $T_1 = t_1$, S is defined on $[t_1, T]$, and

$P(S = t | T_1 = t_1) = g(t)$, $\forall T \geq t \geq t_1$,

where $g(t)$ is not dependent on λ or the sample path of the Poisson process after t_1 .

Condition 2: Given the jumping times of the Poisson process up to any time t , that is, for some

$m \in \mathbb{Z}^+$, given $\{T_1 = t_1, \dots, T_m = t_m, T_{m+1} > t\}$, the distribution of E is

$P[E = t | \{T_1 = t_1, \dots, T_m = t_m, T_{m+1} > t\}] = h_m(t)$, for some $m \in \mathbb{Z}^+$, $\forall T \geq t \geq t_m$,

where $h_m(t)$ is a function independent of λ . Furthermore, given $T_1 = t_1$, the distribution of E is independent of the realization of S :

$P[E = t | T_1 = t_1, S = s] = P[E = t | T_1 = t_1]$, $\forall T \geq t \geq t_1$, $\forall s \geq t_1$.

Now, suppose we observe data which contains for each incident, $S = s$, $E = e$, $X(e) - X(s) =$

M , where $X(s) = m \geq 1$, and all the jumping times within interval $(s, e]$: $JM = \{T_{m+1} =$

$t_{m+1}, \dots, T_{m+M} = t_{m+M}\}$, where $t_{m+1} > s$ and $T_{m+M} \leq e$. Then, conditional on the observed

jumping times before s : $H := \{T_1 = t_1, \dots, T_m = t_m \leq s\}$, the likelihood of the data can be

decomposed as follows:

$P[X(e) - X(s) = M, E = e, S = s, JM | H] = f(e, s, H, JM) p(M | \lambda(e - s))$,

(8)

where $p(M | \lambda(e - s))$ denotes the likelihood of a Poisson distribution with M occurrences and rate parameter $\lambda(e - s)$, and $f(e, s, H, JM)$ is a function that does not depend on λ .

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Expanded PaperInterface specification

```
∀ (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline)
(M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count) (D : M.EndpointDensityPresentation)
{rate : ℝ},
0 < rate →
0 < T.window.exposure →
LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood T M D rate =
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D *
  LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain rate T.window.exposure T.count λ
AppliedModelingLib.Probability.PoissonProcess.RateIndependent fun (_rate : ℝ) =>
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D
```

Retained governing declarations

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow.exposure](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.RateIndependent](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual](#)
- [LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain](#)

Lean-elaborated claim roles

1. parameter: [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
2. parameter: [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count](#)
3. parameter: [M.EndpointDensityPresentation](#)
4. parameter: $\mathbb{R}$
5. assumption: $0 < \text{rate}$
6. assumption: $0 < T.\text{window.exposure}$
7. conclusion: [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood T M D rate =](#)
[LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D *](#)
[LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain rate T.window.exposure T.count λ](#)
[AppliedModelingLib.Probability.PoissonProcess.RateIndependent fun \(_rate : ℝ\) =>](#)
[LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D](#)

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceTheorem1LikelihoodDecomposition

Recorded source-to-Spec screening Verdict: matches

Reason: The target is the source's fixed-history factorization: conditional observation likelihood equals a rate-free factor times the Poisson mass with mean rate times exposure.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 30-41

Verbatim source input

$$p(\tilde{M}_i; \lambda(e_i - s_i)) = \frac{\exp(-\lambda(e_i - s_i))}{\tilde{M}_i!}$$

(2)

[form-feed in source extraction]
is the probability mass function of a Poisson distributed random variable with $\tilde{M}_i$ occurrences and rate parameter $\lambda(e_i - s_i)$, and $f(D_i)$ is a function that does not depend on λ .

[Next verbatim source excerpt]

As reference, the likelihood of a Poisson distribution with $M \geq 0$ occurrences and rate $\lambda(e-s) > 0$ is as follows:
$$p(M | \lambda(e-s)) = \frac{\exp(-\lambda(e-s))}{M!}$$

Expanded PaperInterface specification

$\forall$ (rate exposure : $\mathbb{R}$) (count : $\mathbb{N}$),
0 < rate * exposure $\rightarrow$
LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain rate exposure count =
Real.exp (-(rate * exposure)) * (rate * exposure) ^ count / ↑count.factorial

Retained governing declarations

- [LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{R}$
2. parameter: $\mathbb{R}$
3. parameter: $\mathbb{N}$
4. assumption: 0 < rate * exposure
5. conclusion: LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain rate exposure count =
Real.exp (-(rate * exposure)) * (rate * exposure) ^ count / ↑count.factorial

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceEquation2PoissonCountPMF

Recorded source-to-Spec screening Verdict: matches

Reason: The source and transparent Spec give the same Poisson mass on the explicitly positive rate-times-exposure domain.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 279-288

Verbatim source input

The decomposition in Equation (1) enables standard Poisson rate estimation methods to be applied. For example, the maximum likelihood estimate for λ is:

$$\begin{aligned} \hat{\lambda} &= \bar{y} \\ \hat{\lambda} &= \frac{1}{n} \sum_{i=1}^n y_i \end{aligned}$$

Expanded PaperInterface specification

```
∀ (totalCount : ℕ) (totalExposure : ℝ),
  0 < totalExposure →
    LBG24SpatialUnderreporting.mleRate totalCount totalExposure = ↑totalCount / totalExposure ∧
    (totalCount = 0 →
      ∀ (rate : ℝ),
        0 ≤ rate →
          LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure rate ≤
            LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure
              (LBG24SpatialUnderreporting.mleRate totalCount totalExposure)) ∧
    (totalCount ≠ 0 →
      ∀ (rate : ℝ),
        0 < rate →
          LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure rate ≤
            LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure
              (LBG24SpatialUnderreporting.mleRate totalCount totalExposure))
```

Retained governing declarations

- [LBG24SpatialUnderreporting.mleRate](#)
- [LBG24SpatialUnderreporting.poissonRateLogLikelihood](#)

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℝ
3. assumption: 0 < totalExposure
4. conclusion: LBG24SpatialUnderreporting.mleRate totalCount totalExposure = ↑totalCount / totalExposure ∧
  (totalCount = 0 →
    ∀ (rate : ℝ),
      0 ≤ rate →
        LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure rate ≤
          LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure
            (LBG24SpatialUnderreporting.mleRate totalCount totalExposure)) ∧
  (totalCount ≠ 0 →
    ∀ (rate : ℝ),
      0 < rate →
        LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure rate ≤
          LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure
            (LBG24SpatialUnderreporting.mleRate totalCount totalExposure))
```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceEquation3MaximumLikelihoodEstimate

Recorded source-to-Spec screening Verdict: matches

Reason: The Spec unfolds the aggregate Poisson MLE to totalCount divided by positive totalExposure and formalizes log-likelihood maximization, including the zero-count boundary, exactly as Equation (3) requires.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 832-845

Verbatim source input

First, we assume that the reporting rate is affected by the type vector θ as in a Poisson regression:
for each model, we have a standard reporting rate $\lambda\theta$, defined as:

[U+0001 control character]

$\lambda\theta = \exp \alpha + \beta^T \theta$,

(5)

and so, for each incident i , the likelihood follows from Equation (1) and satisfies the following:

[U+0010 control character]

[U+0011 control character]

$L(\lambda\theta_i | D_i) \propto p(M_i; \lambda\theta_i (e_i - s_i))$,

(6)

13

[form-feed in source extraction]

where β is a coefficient vector of appropriate dimension, (α, β) are coefficients to be estimated, and $p(\cdot)$ is the Poisson likelihood function as defined in Equation (2).

Expanded PaperInterface specification

```
∀ {Feature : Type u_1} [inst : Fintype Feature] (alpha : ℝ) (beta theta : Feature → ℝ)
(T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline)
(M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count)
(D : M.EndpointDensityPresentation),
0 < T.window.exposure →
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood T M D
  (LBG24SpatialUnderreporting.poissonRegressionRate alpha beta theta) =
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D *
  LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain (Real.exp (alpha + ∑ j : Feature, beta j * theta j))
  T.window.exposure T.count Λ
AppliedModelingLib.Probability.PoissonProcess.RateIndependent fun (_rate : ℝ) =>
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D
```

Retained governing declarations

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow.exposure](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.RateIndependent](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual](#)
- [LBG24SpatialUnderreporting.poissonRegressionRate](#)
- [LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: Fintype Feature
3. parameter: ℝ
4. parameter: Feature → ℝ
5. parameter: Feature → ℝ
6. parameter: AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline
7. parameter: LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count
8. parameter: M.EndpointDensityPresentation
9. assumption: 0 < T.window.exposure
10. conclusion: LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood T M D
  (LBG24SpatialUnderreporting.poissonRegressionRate alpha beta theta) =
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D *
  LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain (Real.exp (alpha + ∑ j : Feature, beta j * theta j))
  T.window.exposure T.count Λ
AppliedModelingLib.Probability.PoissonProcess.RateIndependent fun (_rate : ℝ) =>
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D
```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceEquation6PoissonRegressionLikelihood

Recorded source-to-Spec screening Verdict: matches

Reason: The log link is the displayed $\exp(\alpha + \sum_j \beta_j \theta_j)$, and the source's proportional likelihood is exactly a Poisson mass times a rate-independent residual under the selected fixed-history model.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 289-291

Verbatim source input

The proof is in Appendix B.2 and follows from decomposing the likelihood function into full conditional probabilities. The key step is to find the conditions under which one can use data-dependent observation period length $E_i - S_i$, while maintaining a Poisson likelihood for reports within the period.⁵ The result extends to the zero-inflated model we use in our empirical estimation.

[Next verbatim source excerpt]

We further fit a zero-inflated Poisson regression [19], which corresponds to a data generating process in which, independently for each incident, only one report can be generated with probability γ , and thus there are 0 duplicate reports: $\tilde{M}_i = 0.9$ For example, perhaps those incidents are personal requests for service (e.g., to plant a tree somewhere or to prune a tree to enable certain construction), for which duplicates are unlikely; alternatively, officials may physically mark the location of an incident as “report received; full inspection pending,” preventing future reports. Under this case, the likelihood of the data satisfies:

[U+0011 control character]

[U+0010 control character]

(
 $\tilde{M}_i \geq 1;$
 $(1 - \gamma)^p \tilde{M}_i; \lambda \theta_i (e_i - s_i),$
 $L(\lambda \theta_i | D_i) \propto$
(7)

$\gamma + (1 - \gamma)^p (0; \lambda \theta_i (e_i - s_i)), \tilde{M}_i = 0.$

Expanded PaperInterface specification

```
∀ (M : LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel),
  LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood M.gamma M.rate M.exposure 0 =
    M.gamma + (1 - M.gamma) * LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain M.rate M.exposure 0 ∧
  (∀ (count : ℕ),
    1 ≤ count →
      LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood M.gamma M.rate M.exposure count =
        (1 - M.gamma) * LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain M.rate M.exposure count) ∧
  ∀ {Incident : Type u_1} (incidents : Finset Incident)
    (model : Incident → LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel) (count : Incident → ℕ),
    LBG24SpatialUnderreporting.zeroInflatedIndependentIncidentFamilyLikelihood incidents model count =
      ∏ i ∈ incidents,
        LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood (model i).gamma (model i).rate (model i).exposure
        (count i)
```

Retained governing declarations

- [LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel](#)
- [LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain](#)
- [LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood](#)
- [LBG24SpatialUnderreporting.zeroInflatedIndependentIncidentFamilyLikelihood](#)

Lean-elaborated claim roles

```
1. parameter: LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel
2. conclusion: LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood M.gamma M.rate M.exposure 0 =
  M.gamma + (1 - M.gamma) * LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain M.rate M.exposure 0 ∧
  (∀ (count : ℕ),
    1 ≤ count →
      LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood M.gamma M.rate M.exposure count =
        (1 - M.gamma) * LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain M.rate M.exposure count) ∧
  ∀ {Incident : Type u_1} (incidents : Finset Incident)
    (model : Incident → LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel) (count : Incident → ℕ),
    LBG24SpatialUnderreporting.zeroInflatedIndependentIncidentFamilyLikelihood incidents model count =
      ∏ i ∈ incidents,
        LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood (model i).gamma (model i).rate (model i).exposure
        (count i)
```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceZeroInflatedLikelihoodExtension

Recorded source-to-Spec screening Verdict: matches

Reason: The Spec gives the source’s gamma structural-zero plus 1-gamma Poisson branches and its independent-incidents observed likelihood product.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 1453-1520

Verbatim source input

Lemma 1. Suppose each incident gets reported independently, and the distribution of the interval of reporting duration T_i has density function $f(\cdot) : [0, \infty) \rightarrow \mathbb{R}$. Then under steady state, $N_{\theta}^{\text{observed}}$ follows a Poisson process with rate Λ_{θ} , where

[U+0014 control character]
[U+0012 control character] $\int_0^{\infty}$
[U+0013 control character]
[U+0015 control character]

$$\Lambda_{\theta} = \Lambda_{\theta} \int_0^{\infty} \exp(-\lambda_{\theta} u) du f(t) dt.$$

0

In the simplest time-homogeneous case, this rate simplifies to:

[U+0014 control character]
[U+0015 control character] $\int_0^{\infty}$
[U+0013 control character]

$$\Lambda_{\theta} = \Lambda_{\theta} \int_0^{\infty} \exp(-\lambda_{\theta} t) f(t) dt.$$

Proof of Lemma 1. Let $m(t)$ be the number of times an incident is reported in an interval of t , starting from its birth. We know from the model assumption that $m(t)$ follows a Poisson distribution:

[U+0012 control character] $\int_0^{\infty}$
[U+0013 control character]
 $m(t) \sim \text{Poisson}$
 $\lambda_{\theta} \int_0^{\infty} du$
0

23

[form-feed in source extraction]

In steady state, each unique incident gets reported with probability

$\int_0^{\infty}$
 $\Pr[m(t) \geq 1 | T_i = t] f(t) dt$
 $p =$
0
[U+0013 control character][U+0015 control character]
[U+0012 control character] $\int_0^{\infty}$
 $\int_0^{\infty} \lambda_{\theta} \int_0^{\infty} du f(t) dt$
 $=$
 $1 - \exp(-\lambda_{\theta} t)$
0
0
[U+0013 control character]
[U+0012 control character] $\int_0^{\infty}$
 $\lambda_{\theta} \int_0^{\infty} du f(t) dt,$
 $\exp(-\lambda_{\theta} t)$
 $= 1 -$
0
0

which simplifies under time-homogeneity to

$$\int_0^{\infty} \exp(-\lambda_{\theta} t) f(t) dt.$$

$p = 1 -$
0

Over a time interval of t , the total number of incidents of type θ that happen, $N_{\theta}(t)$ follows a Poisson process with rate Λ_{θ} . Conditional on $N_{\theta}(t) = n$, $n = 0, 1, \dots$, under steady state, $N_{\theta}^{\text{observed}}(t)$ follows a binomial distribution with parameters (n, p) . Thus $N_{\theta}^{\text{observed}}(t)$ follows a Poisson distribution with rate $\Lambda_{\theta} p$, which completes the proof.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Expanded PaperInterface specification

$\forall \{\Omega : \text{Type } u_1\} [\text{inst} : \text{MeasurableSpace } \Omega] \{P : \text{MeasureTheory.Measure } \Omega\}$
(M : LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P),
($\forall (t : \mathbb{N}\text{Real}), \text{Measurable } (M.\text{calendarFirstReports}.\text{calendarCount } t)) \wedge$
($\forall^m (\omega : \Omega) \partial P, M.\text{calendarFirstReports}.\text{calendarCount } 0 \ \omega = 0$) $\wedge$
($\forall^m (\omega : \Omega) \partial P, \text{Monotone fun } (t : \mathbb{N}\text{Real}) \Rightarrow M.\text{calendarFirstReports}.\text{calendarCount } t \ \omega$) $\wedge$
ProbabilityTheory.HasIndepIncrements M.calendarFirstReports.calendarCount P $\wedge$
 $\forall \{s t : \mathbb{N}\text{Real}\},$
 $s \leq t \rightarrow$
 $\forall (\text{observedCount} : \mathbb{N}),$


```

P.real
{omega : Omega |
M.calendarFirstReports.calendarCount t omega - M.calendarFirstReports.calendarCount s omega =
observedCount} =
AppliedModelingLib.Probability.PoissonProcess.countLikelihood
(LBG24SpatialUnderreporting.continuousDurationObservedIncidentRateOfCumulativeIntensity
M.calendarFirstReports.incidentRate (fun (u : ℝ) => ∫ (v : ℝ) in 0..u, M.reportingIntensity v)
M.durationDensity)
(↑t - ↑s) observedCount

```

Retained governing declarations

- [AppliedModelingLib.Probability.PoissonProcess.countLikelihood](#)
- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel](#)
- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.calendarCount](#)
- [LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel](#)
- [LBG24SpatialUnderreporting.continuousDurationObservedIncidentRateOfCumulativeIntensity](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: MeasurableSpace Omega
3. parameter: MeasureTheory.Measure Omega
4. parameter: LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Omega P
5. conclusion: (∀ (t : NNReal), Measurable (M.calendarFirstReports.calendarCount t)) ∧
(∀m (omega : Omega) ∂P, M.calendarFirstReports.calendarCount 0 omega = 0) ∧
(∀m (omega : Omega) ∂P, Monotone fun (t : NNReal) => M.calendarFirstReports.calendarCount t omega) ∧
ProbabilityTheory.HasIndepIncements M.calendarFirstReports.calendarCount P ∧
∀ {s t : NNReal},
s ≤ t →
∀ (observedCount : ℕ),
P.real
{omega : Omega |
M.calendarFirstReports.calendarCount t omega - M.calendarFirstReports.calendarCount s omega =
observedCount} =
AppliedModelingLib.Probability.PoissonProcess.countLikelihood
(LBG24SpatialUnderreporting.continuousDurationObservedIncidentRateOfCumulativeIntensity
M.calendarFirstReports.incidentRate (fun (u : ℝ) => ∫ (v : ℝ) in 0..u, M.reportingIntensity v)
M.durationDensity)
(↑t - ↑s) observedCount

```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceLemma1CalendarTimeDurationObservedProcess

Recorded source-to-Spec screening Verdict: matches

Reason: Lemma 1 states that calendar-time observed incidents form a Poisson process at incidentRate times the duration-mixture reporting probability; under the approved first-report calendar interpretation, the target gives measurable monotone counts, independent increments, and the exact Poisson increment law at that rate.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 1433-1452

Verbatim source input

Proposition 1. Consider the simplest setting: both incident occurrence and reporting follow timehomogeneous Poisson processes, with $\Lambda\theta(\tau) = \Lambda\theta$ and $\lambda\theta \hookrightarrow (\tau) = \lambda\theta$. Further, let incident reporting duration T_i be distributed according to F . Then using just the number of observed incidents $N\theta_{\text{observed}}(T)$ of type θ in a known time duration $[0, T]$, the reporting rate $\lambda\theta$ is not identifiable. In other words, $N\theta_{\text{observed}}(T)$ is a function of both $\Lambda\theta$ and $\lambda\theta$:

$N\theta_{\text{observed}}(T)$

$= \Lambda'\theta$

$T \rightarrow \infty$

T

[U+0014 control character]

[U+0015 control character]

$Z \infty$

$\exp(-\lambda\theta t) dF(t)$.

$\Lambda\theta = \Lambda\theta 1 -$

$\lim$

where

0

The proof of Proposition 1 follows directly from the following Lemma.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Additional assumptions The Proposition 1 nonidentifiability construction is checked on the nondegenerate observed-count-rate branch $0 < \text{observedRate}$.

Expanded PaperInterface specification

```
∀ {Omega1 : Type u_1} {Omega2 : Type u_2} [inst : MeasurableSpace Omega1] [inst_1 : MeasurableSpace Omega2]
{P1 : MeasureTheory.Measure Omega1} {P2 : MeasureTheory.Measure Omega2}
(M1 : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Omega1 P1)
(M2 : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Omega2 P2) (observedRate : ℝ),
0 < observedRate →
M1.reportingRate ≠ M2.reportingRate →
M1.durationDensity = M2.durationDensity →
0 < LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M1.reportingRate M1.durationDensity →
0 < LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M2.reportingRate M2.durationDensity →
M1.calendarFirstReports.incidentRate =
  observedRate /
  LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M1.reportingRate
  M1.durationDensity →
M2.calendarFirstReports.incidentRate =
  observedRate /
  LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M2.reportingRate
  M2.durationDensity →
(∀m (omega : Omega1) ∂P1,
  Filter.Tendsto (fun (n : ℕ) => ↑(M1.observedUniqueIncidentCount n omega) / ↑n) Filter.atTop
  (nhds observedRate)) ∧
(∀m (omega : Omega2) ∂P2,
  Filter.Tendsto (fun (n : ℕ) => ↑(M2.observedUniqueIncidentCount n omega) / ↑n) Filter.atTop
  (nhds observedRate)) ∧
M1.reportingRate ≠ M2.reportingRate ∧
∃ (observedProcess1 :
  AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw
  Omega1 P1)
(observedProcess2 :
  AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw
  Omega2 P2),
observedProcess1.count = M1.calendarFirstReports.calendarCount ∧
observedProcess2.count = M2.calendarFirstReports.calendarCount ∧
observedProcess1.rate = observedRate ∧ observedProcess2.rate = observedRate
```

Retained governing declarations

- [AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw](#)
- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel](#)
- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.calendarCount](#)
- [LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel](#)
- [LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel.observedUniqueIncidentCount](#)
- [LBG24SpatialUnderreporting.continuousDurationFirstReportProbability](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: Type u_2
3. parameter: MeasurableSpace Omega1
4. parameter: MeasurableSpace Omega2
5. parameter: MeasureTheory.Measure Omega1
6. parameter: MeasureTheory.Measure Omega2
7. parameter: LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Omega1 P1
8. parameter: LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Omega2 P2
9. parameter: ℝ
10. assumption: 0 < observedRate
11. assumption: M1.reportingRate ≠ M2.reportingRate
12. assumption: M1.durationDensity = M2.durationDensity
13. assumption: 0 < LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M1.reportingRate M1.durationDensity
14. assumption: 0 < LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M2.reportingRate M2.durationDensity
15. assumption: M1.calendarFirstReports.incidentRate =
  observedRate / LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M1.reportingRate M1.durationDensity
16. assumption: M2.calendarFirstReports.incidentRate =
  observedRate / LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M2.reportingRate M2.durationDensity
17. conclusion: (∀m (omega : Omega1) ∂P1,
  Filter.Tendsto (fun (n : ℕ) => ↑(M1.observedUniqueIncidentCount n omega) / ↑n) Filter.atTop (nhds observedRate)) ∧
  (∀m (omega : Omega2) ∂P2,
  Filter.Tendsto (fun (n : ℕ) => ↑(M2.observedUniqueIncidentCount n omega) / ↑n) Filter.atTop (nhds observedRate)) ∧
  M1.reportingRate ≠ M2.reportingRate ∧
  ∃ (observedProcess1 :
    AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Omega1 P1)
    (observedProcess2 :
    AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Omega2 P2),
  observedProcess1.count = M1.calendarFirstReports.calendarCount ∧
  observedProcess2.count = M2.calendarFirstReports.calendarCount ∧
  observedProcess1.rate = observedRate ∧ observedProcess2.rate = observedRate
```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceProposition1CalendarTimeNonidentifiability

Recorded source-to-Spec screening Verdict: matches

Reason: For each distinct reporting rate, the Spec fixes the compensating incident rate as observedRate divided by the displayed first-report probability $1 - \int \exp(-\lambda t) dF(t)$. The selected calendar-count declarations then give both the common-rate homogeneous Poisson process and the stated almost-sure count-rate limit.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation